

You must show work to receive credit. Each problem is worth 10 pts unless stated otherwise.

Use the given conditions to write an equation for the line in slope-intercept form, if possible.

1. Slope = 5, passing through $(1, 3)$.

2. Through $(2, -3)$ and $(-3, 4)$.

Write a cost function for the problem. Assume that the relationship is linear.

3. Alfred owns a small publishing company. Printing a book has a fixed cost of \$500 plus \$3 per book printed. Find the cost function for printing x number of books.

Write a revenue function for the problem.

4. If Alfred's publishing company sells books for \$12 a piece, find the revenue function for selling x number of books.

Solve the problem.

5. Producing x units of tacos costs $C(x) = 5x + 20$; revenue is $R(x) = 15x$, where $C(x)$ and $R(x)$ are in dollars.
 - a. Find the profit function for selling x number of tacos.

- b. How many tacos must be sold to make a profit of \$120?

6. Suppose that the supply and demand functions for honey is

$$p = S(q) = 0.3q + 2.7 \text{ and } p = D(q) = 6.9 - 0.4q$$

where p is the price in dollars for an 8-oz container and q is the number of barrels.

- a. Find the equilibrium quantity.

- b. Find the equilibrium price.

Find the equation of the least squares line.

7. The total amount of consumer credit has been increasing in recent years. The following tables gives the total US outstanding consumer credit in billions of dollars for selected years.

Year	2004	2005	2006	2007	2008
Consumer Credit	2219.5	2319.5	2415.0	2551.9	2592.1

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

8. [5 points] Explain what it means to “break even” in problem 5.
9. [5 points] The correlation coefficient from problem 7 is $r \approx 0.991$. Does this indicate a good fit of the least squares? If another data set has a correlation coefficient of $r = -.999$, does the data fit more or less closely to a line than the data from problem 7?